

accuracy. And the best part? They use very little power and are highly reliable. Can you see how this helps industries that depend on precision?

Robots also get tactile sensors that allow them to handle objects with care. These sensors detect force and let robots adjust their grip, making them more adaptable and efficient. Ever wonder how robots can hold delicate objects or know when something slips? Yes, it is these sensors that make it possible.

TMR sensors in data centres

So, how do data centres manage the increasing power demands efficiently? The key is to have power supplies that are both energy-efficient and cost-effective. Managing the heat from continuous high power consumption becomes a real challenge, especially with the explosion of connected devices and artificial intelligence (AI) workloads. But here is where advanced current sensing technology comes in, helping data centres monitor and manage power flow with precision, ensuring energy is used efficiently without wasting it.

Unlike older technologies like Hall and giant magnetoresistance (GMR), TMR sensors provide higher sensitivity, faster response times, and consume less power. They detect magnetic fields created by electrical currents and convert them into precise voltage outputs, so you can monitor power flow in real time, without interrupting it.

The best part? TMR sensors support high-speed switching and low-noise performance—crucial for the high demands of AI and internet of things (IoT) systems. With bandwidths reaching up to 1MHz and response times as quick as 300 nanoseconds, they are perfect for managing power in environments that require rapid power conversion.

Plus, they are compact, scalable, and cost-effective, making them ideal for applications where power density is high and power consumption needs to be low.

But wait, there is more. TMR sensors provide accurate current measurements across a range of conditions, including high temperatures, and can support power applications from 80A to over 800A. As we move towards an increasingly connected world, the need for efficient power management will only grow. TMR sensors are already making a significant difference by reducing power waste and improving efficiency in data centres and AI environments. By embracing TMR technology, data centres can meet the demands of the connected world while staying sustainable.

How TMR sensors protect GaN and SiC devices

As wide bandgap (WBG) semiconductors like gallium nitride (GaN) and silicon carbide (SiC) become more common in electronics, there is a challenge we all face: the need for precise, high-speed sensing. These materials are appearing in everything from automotive inverters to renewable energy power converters. So, what do we do to ensure these high-performance transistors operate at their best? The answer lies in advanced sensing technology that manages power and temperature, keeping things running smoothly and protecting sensitive components.

But how do we keep these powerful devices safe? Well, that is where TMR sensors come in. These sensors are becoming crucial in WBG applications, helping monitor both current and temperature, and ensuring that the devices last longer and operate safely.

Let us dive into why GaN and SiC transistors are so special. These

materials can handle high voltages and switch at fast speeds, making them perfect for high-demand applications like photovoltaic inverters, power adaptors, and electric vehicle chargers. But here is the thing: the same features that make these devices so efficient also make them sensitive to quick changes in circuit conditions, especially when there is a sudden shift in current and temperature during fast switching from gate drivers.

Now, what happens if we do not manage these rapid changes? It can lead to overheating and even failure of the switches. However, TMR sensors are here to help. Unlike traditional sensing solutions, like shunt resistors and amplifiers, TMR sensors keep up with the high-speed needs of GaN and SiC applications.

What is even better? The latest generation of TMR sensors is specifically designed for high-speed, high-power WBG applications. They offer high bandwidth and accuracy without sacrificing performance. For example, these sensors can handle frequencies up to 5MHz, making them perfect for efficient power conversion in modern systems.

And let us talk about space. With devices getting smaller and more powerful, space becomes more critical. TMR sensors are suitable—they have a small footprint on printed circuit boards (PCBs), freeing up space for other components while still delivering excellent performance. Plus, their low power consumption makes them a smart choice for designs that balance performance with cost. **EFY**

This article, inspired by a conversation with Ram Sathappan, Senior Director of Marketing at Allegro MicroSystems, is authored by Nidhi Agarwal, a journalist at EFY and an electronics and communication engineer with over five years of academic experience, who specialises in development boards and IoT cloud solutions.